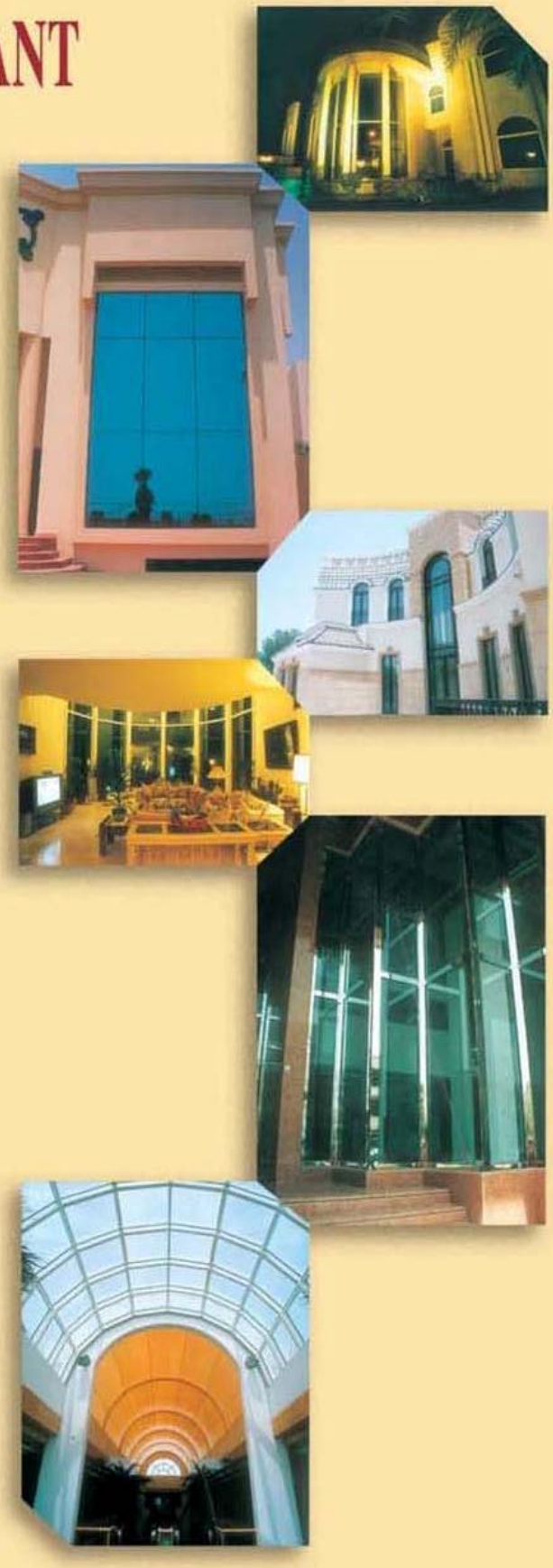
AL JAZIRAH ALUMINIUM PLANT



Al Jazirah ...

Name with Deep Roots

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Introduction

Al Jazirah Aluminium Plant
For Doors & Windows Systems

INTRODUCTION

AL-JAZIRAH is one of the largest, most reputable Saudi companies in the field of designing, manufacturing and installing, of aluminum structural glazing windows, doors, curtain walls, skylights, cladding, to serve the Saudi construction industry.

The AL-JAZIRAH operation has started early 1975, and throughout the years expanded tremendously to add:

*Upstream production lines enabling more control on raw material prices & Quality with ISO 9001:2000 certified.

* Horizontal expansion with new products and options to expand its market share.

Now a days, AL-JAZIRAH is, by far, are one of the market leader and has proved throw the years its leadership over all its eventual competitors.

THE FACTORY

The total factory area:

4200 (Four thousand two hundreds) M2

The total builds up area:

3900 (Three thousands nine hundreds) M2

The administrative building area:

700 (Seven hundreds) M2

The Representation Al-khobar show room area:

200 (Two hundreds) M2

The factory consists of several production lines:

First: The Production of Windows, Doors & Curtain Walls, consist of 5 main fabrication Facilities:

1- Standard Windows Fabrication lines.
2- Non Standard Windows Fabrication lines.
3- Curtain wall, Structural Glazing, Skylight, and special window and door design fabrication line.
4- Auxiliary Products Fabrication lines.
5- Hardware Assembly.

Total Productivity of 60,000 m^2 / year

The following production tools are available:

1- Sawing machines.
2- Milling and drilling machines.
3- Corner assembly machines.
4- Bending machines.
5- Quality control supervisors and inspection according to the international Standard specifications.

There are also the raw material stores that are designed to protect the mill-finished aluminum from the external environmental effects and to organize the stock according to the most advanced storage systems. In addition to the raw material stores, there are the finished product stores where specialized technicians pack the finished product after being approved by the quality control supervisors and prepare it for mobilization and installation.

THE TECHNICAL DEPARTMENT:

First: The Design Office.

Second: The Research and Development Department (R&D).

First: The Design Office.

AL-JAZIRAH depends upon its design office in studying and designing all its projects where architects and engineers design the architectural openings and study each project separately, using the available facilities that are summarized as follows:

* 2 Network Servers.
* 12 PCs, 3 Laptops.
* 3 Laser Printers, 3 DeskJet Printers
* 3 Photocopier cum Fax Machines size A3, A4.

The design office staff consists of 5 architects and engineers of high qualification and experience in architectural aluminum works. They are classified as follows:

* 2 supervising engineers.
* 2 design engineers.
* 1 newly graduate engineers.

Second: The Research and Development Department (R &D):

In addition to the design office and its facilities, AL-JAZIRAH has a research and development department, where the latest technology in the field is used to check and develop the current systems under production and modifies them, so as to suit the toughest international standards. Customers' needs and special designs are considered at the R & D department and answers to many technical and field problems are always given.

The R&D department also trains AL-JAZIRAH's new staff with an appropriate training program.

HUMAN RESOURCES:

The number of permanent employees are more than 60 spread between the following departments:

Marketing & Sales Department

4 Engineers and Assistant Technicians

Technical Department

52 Engineers, Technical Supervisors and Labors

Financial & Administrative Department

4 Accountants & Administrators

The company employee's qualifications:

* Master Degree: 2
* University Degree: 6
* Technical & Secondary School: 37
* Below Secondary School: 15

CONTACT US

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General Specification

General Specification

ISO Quality Cretified

AL JAZIRAH SWISS

مصنع المنيوم الجزيرة

AL-JAZIRAH ALUMINIUM PLANT

الخصائص العامة :

١. الإبداع في تصميم القطاعات على الطراز السويسري وذلك بالمحافظة على الجودة العالية تحت أقل التكاليف .
٢. تصميم القطاعات يمتاز بوجود الأعصاب الهندسية للمتانة والقوة .
٣. الأقفال والمسكات والفصليات والإكسسوارات أوربية أمريكية أصلية الصنع .
٤. وجود مقاطع خاصة ذات سماكات عالية تتراوح مابين ٢,٢ مم فما فوق وذلك للمتانة والقوة .
٥. استخدام براغي ستانلس استيل لمنع التآكل والصدأ .
٦. التجميع يمتاز بالبراغي المخفية لمزيد من الأناقة والجمال .
٧. تفادي وجود مسامير في الاسفل لمنع أي تهريبات .
٨. تخصيص جودة عالية من السيلكون عند التركيب وذلك لمقاومة التشقق والحرارة .
٩. إمكانية إضافة الحواجب أو البراويز الديكورية لإضافة لمسة من الفخامة .
١٠. مواد شبك الناموس مصنعة من الألمنيوم الخالص لمنع التآكل والصدأ .
١١. نظام النوافذ المانع لاهتزازات درف شبك الناموس المضاد للتيارات الهوائية .
١٢. نظام النوافذ ذو الفرش المزدوجة لمنع دخول الغبار أو الحشرات .
١٣. السهولة والمرونة في صيانة وتغيير الزجاج دون الحاجة إلى فك الدرفة .
١٤. زاوية تجميع مخفية مضغوطة بالهيدروليك تمتاز بالصلابة والقوة ولمنع التسربات .
١٥. تزويد الأبواب بالكعب العريض لتحمل الإرتطامات ولتناسق الفريد .
١٦. توفرة كافة الوان الدهان العادي والخشابي بشتى أنواعه .
١٧. جميع الربلات من نوع " إي . بي . دي . إيم " المقاومة للظروف الصحراوية .
١٨. وجود خاصية الزجاج الأوروبي الأمريكي المزدوج (المعالج حراريا) مع مختلف الأنواع والسماكات .
١٩. إعتماد شركات سحب عالية الجودة (الوبكو) ومعتمدة بمعايير "اللويز" العالمية .
٢٠. الضمان الشامل ضد الصوت والهواء والماء والغبار .

General Specifications :

1 - Made from heavy-duty design following Swiss Standards but economical.
2 - Extra strength through braces and beads in profile design.
3 - European and American locks, handle, hinges and other accessories.
4 - Aluminum wall thickness ranging from 2.2 mm and above for strength & quality.
5 - All screws stainless steel to avoid rusting.
6 - No visible screws for aesthetic purposes.
7 - Prohibition of any screw at sill to prevent leakages.
8 - High quality silicon sealant during installation that resist crack and heat.
9 - Possibility of addition of flashings and decorative profiles for aesthetic purposes.
10 - Aluminium fly screen mesh to prevent rust.
11 - Special design to avoid fly screen vibration against wind pressure.
12 - Hi-fin-system dual brushes to avoid any dust or insect penetration.
13 - Possibility to change the glass without dismantling the shutters.
14 - Heavy duty concealed corner cleats that crimped hydraulically for more sturdiness.
15 - High Door Kick plate for more strength & beauty.
16 - Availability of any powder coated color, including wooden shades.
17 - All gaskets from EPDM material that can resist Desert climate.
18 - Double glazed European and American glass (Temp.) of all types in various thickness.
19 - Approved high quality extrusion companies (ALUPCO) complying with standard "ALLOYS"
20 - Fully guaranteed against sound, wind, water and dust penetration.

P.O.2166 Dammam 31451 - 1st Industrial - Saudi Arabia ص.ب. ٢١٦٦ الدمام ٣١٤٥١ المدينة الصناعية الاولى

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GENERAL SPECIFICATIONS
FOR
(ALUMINUM WINDOWS, DOORS AND CURTAIN WALLS)

Systems have certain codes related to profiles used

Hinged systems

- **AL JAZIRAH AJ 45** — Indicates heavy weight hinged system profiles 2.2 Mm thickness, 45 mm frame depth including Turn-tilt, tilt-slide and folding windows

- **AL JAZIRAH AJ 45 R** — Indicates an additional heavy duty weight hinged system profiles 3.2 Mm thickness, 45 mm frame depth

Sliding systems

- **AL JAZIRAH AJ 32** — Indicates heavy weight sliding system profiles 2.2 Mm thickness, 100 mm frame depth

- **AL JAZIRAH AJ 40** — Indicates an additional heavy duty weight sliding system profiles 3.0 Mm thickness, 100 mm frame depth

Curtain wall systems

- **AL JAZIRAH C.W.** — Indicates conventional curtain wall system

- **AL JAZIRAH S.G.CW** — Indicates structural glazing curtain wall system

All sections & profiles are made of extruded aluminum alloys free from defects impairing strength or appearance.

Aluminum alloy is 6063 – T6 which has an ultimate tensile strength of 15 kg/ mm2 & hardness (BRINELL) of 60 kg / mm2, according to DIN BS 1474.

FABRICATION & ASSEMBLY:

* All corners are mechanically joined using heavy aluminum angles.
* Stainless steel screws are used for hardware fixation.
* A drainage system is provided for rain and cleaning water.
* Hinged doors & windows are provided with three gaskets to prevent air & dust, Infiltration & improve sound insulation.

These gaskets are:

1- Central gaskets.
2- Lateral gaskets.
3- Glass beading gaskets.

* Sliding doors & windows are provided with glass beading rubber gasket, heavy Duty brushes and central dust plug.
* All rubber gaskets are of pure EPDM. (See attached technical specification for Rubber gaskets).
* Special design of AL-JAZIRAH profiles gives the flexibility of using different Sizes of glazing:

** From 6 mm to 10 mm thickness for single glazing.
** From 18 mm to 24 mm thickness for double-glazing.
** From 6.8 mm to 26 mm thickness for laminated glass.

SURFACE FINISHING:

All profiles are powder-coated electro statically at ALUPCO powder coating plant.

(See AL-JAZIRAH electrostatic powders coating detailed specifications)
Or by using the Anodizing surface finish process .

ACCESSORIES & FITTINGS:

Heavy duty, high quality European /American made accessories and fittings are used on all AL-JAZIRAH products. They give a large choice to select the suitable way of opening such as Turn-Tilt, Tilt-Slide, Pivoted, Folding...etc.

Special locking mechanism with locking points all around the frame perimeter ensures complete tightness and security.

Heavy duty Polyamide Rollers with Needle-Bearing are used for sliding windows thus to enable the smooth movement of large sliding sashes (up to 3.2m x 2.2m for each sliding sash).

INSTALLATION:

Due to the design of AL-JAZIRAH heavy weight aluminum frames it can be fixed directly on concrete or plastered walls without any sub-frames.

* AL-JAZIRAH can provide steel or aluminum sub-frames with standard dimension of 2x5 cm if required.

PERFORMANCE REQUIREMENTS:

All aluminum systems performance requirements comply with BS 6365 Part 1, DIN 18055, 4108 and 4109.

1) AIR INFILTRATION:

Cubic m/ hr per meter length of visible perimeter of glass at a pressure Difference of 200 Pa (N/ m2).

Classification: A3 (Reinforced)

2) WATER PENETRATION:

No water leakage at a testing pressure of 550 Pa (N/ m2).

Classification: EE (Exceptional)

3) DEFLECTION:

Members retaining an insulating glass unit shall not deflect more than L/175 over length or height of the glass unit when submitted to the design wind pressure and allowable factor of safety.

4) THERMAL INSULATION:

Maintain DIN 4108 requirements.

5) SOUND INSULATION:

Maintain DIN 4109 sound proofing requirements.

Special combinations of window, door frames or curtain walls using sound insulating glass can achieve a Sound Reduction index Rw of 49 dB.

TECHNICAL CRITERIA OF A WINDOWS

As per French Standards (NFP 20.302, NFP 20.501)

1) AIR INFILTRATION (m3/h.m2, or m3/h.m):

Measuring the quantity of air discharged from a unit under a certain external / internal pressure difference, in view of the unit area or of the unit perimeter. (See attached charts).

Class A1 = Normal.

Class A2 = Improved.

Class A3 = Reinforced.

2) WATER PENETRATION:

Measuring the maximum pressure PE under which the unit remains tight, in a test imitating the rain effect.

Class E1 = Normal.	50 Pa < PE < 150 Pa
Class E2 = Improved.	150 Pa < PE < 300 Pa
Class E3 = Reinforced.	300 Pa < PE < 500 Pa
Class EE = Exceptional.	PE > 500 Pa

3) MECHANICAL RESISTANCE:

* Deformation Under Loads Imitating Wind Pressure Effect:

Under a pressure of 500 Pa (or 1000 Pa), maximum deflection of any element of the unit (apart from the glass) should not exceed 1/200 of its span.

Class V1	500 Pa
Class V2	1000 Pa

* Sudden Pressure Effect:

Under the stated pressures, the unit should not break nor open suddenly.

Class V1	900 Pa
Class V2 1700 Pa	
Class VE (Exceptional) >	2300 Pa

RUBBER GASKETS

Tightness of all operable sashes (hinged), and of glazing is assured by EPDM gaskets.

EPDM (Ethylene Propylene Diene Monomer) is a rubber mixture extremely resistant to severe weather changes. Acids, alkaline, Ozone, salts, heat, cold, nor moisture does not affect it.

Specifications of rubber gaskets can be as follows:

* The gaskets' material shall consist of at least 50% by weight of basic rubber Hydrocarbon.
* The material shall not contain any crude or reclaimed rubber.
* The material shall be homogeneous, and free from defects.
* Cured EPDM shall meet the following properties when tested in accordance With ASTM test slabs and compression set plugs:

Property	**Test Procedure**	**Typical Values**
Hardness (±5)	ASTM D - 2000	50
Tensile Strength (N/ sq. mm)	ASTM D - 2000	11.2 ® 12.4
Elongation (%)	ASTM D - 412	400
Heat Aging	ASTM D - 865	Complies

THERMAL INSULATING GLASS

One of the facilities AL-JAZIRAH is providing to its clients is the thermal insulating glass Products.

GENERAL

Thermal insulating glass consists of two panes of glass fixed at a given distance apart by a spacer and hermetically sealed at their periphery. The air space so formed is filled with dehydrated air (or other gases). Airspace widths vary from 6 mm to 16 mm. Units can be manufactured from any of the range of flat glasses: annealed, toughened or laminated.

Solar control units may be manufactured incorporating the full range of body tinted, surface modified and surface coated (reflective) glass as well as by incorporating a thin metallic deposition overall on a glass surface adjacent to the airspace.

CONSTRUCTION:

AL-JAZIRAH Thermal Insulating glass is composed of two glass panes, separated by a space filled with perfectly dehydrated air or gas, held in place by a perimeter metal spacer, which is filled with desiccant. The whole unit is hermetically assembled by a double barrier sealant.

Thicknesses of the spacer available are 6 mm, 9 mm, 10 mm and 16 mm.

TYPES OF THERMAL INSULATING GLASS:

Thermal insulating glass is available in a large number of combinations, depending on the thickness and shade of the glass, on the thickness of the spacer, and on the type of the airspace (air or gas).

Because of the different manufacturing techniques available and the large range of possible glass combinations, AL-JAZIRAH should be consulted for detailed information.

THICKNESS AND SIZE:

The most common thicknesses (of the complete glass unit) are from 18 to 24 mm, but any inquiry may be considered.

The maximum dimensions are those of the glass panes, i.e. 3200 mm x 2250 mm.

PERFORMANCE:

The first function of insulating glass is to reduce the thermal exchange between the two environments that it separates, which brings many advantages:

- Reduction of Installation cost for air-conditioning and/or heating.
- Lower energy consumption.
- Better comfort through reduction of disagreeable drafts near the windows.
- Possibility of using bigger glazed areas for a given room temperature without Increasing the energy consumption.

The insulating power of a wall, i.e. the quantity of energy which passes through it under well-defined conditions, is given by its k value or U value, and is expressed in W/sqm .K, following ISO DP 9050 standard or in Btu/hr/sqft, following ASHRAE recommendations.

A normal single glazing of whatever thickness, has a U or k value (ISO) in the order of 5.8 W/sqm .K., whereas an insulating glazing composed of the same glass with an airspace of 12 mm has a U or k value of around 2.8 W/sqm .K., which means that in the last case the heat loss is less than half.

There are means to further improve the insulating power or to add other functions to the insulating unit such as solar protection, safety, and acoustical insulation.

LAMINATED GLASS

One of the facilities AL-JAZIRAH is providing to its clients is the laminated glass products.

GENERAL

Laminated glass consists of two or more panes of glass with an interlayer of reinforcing material between each two panes. The interlayer are permanently bonded to the glass panes under carefully controlled processing. The interlayer are resistant to penetration and they absorb impact shock, holding the glass in place and preventing extensive spalling of glass fragments. The degree of protection provided by laminated glass depends on the overall thickness and construction.

Laminated glass can usually be cut to size and worked after manufacture; suitable techniques depend on the type and construction of the laminated glass.

CONSTRUCTION:

Laminated glass may be used for a wide range of functions; hence it is necessary to define the type and thickness of the glass and the requirements for the interlayer. The number of layers commonly refers to laminated glasses, i.e. two panes of glass with one interlayer is known as three-ply.

Thicknesses of the interlayer available are 0.8 mm, 1.0 mm and 1.2 mm.

Tests on laminated glass has shown the following results:

Accelerated ageing has been performed on 4/0.8/4 laminates. The following results were obtained:

Q-panel (UV test):	very slight yellowing after 2400 hours.
Klima	30 days, no change.
Hot water test:	14 days, no change.

Description of the tests used:

Q-panel (ASTM G 53-7):

12-hour cycle:	8 hours UV -A light (310-390 NM) Black panel temperature: 60 OC
	4 hours 100% humidity Panel temperature: 40 OC

Klima

24 hour cycle:	16 hours at + 90 OC 8 hours at - 25 OC
Hot water test:	immersion in water at 90 OC.

Impact properties of glass laminates

Based laminate satisfies the DIN 52338 requirements for building applications:
1,040 g falling ball, 2.5 m height, 1 impact.

TYPES OF LAMINATED GLASS:

Laminated glass is available in a large number of combinations, depending on the thickness and shade of both the glass and interlayer, and on the type of glass (annealed, tempered, figured, rolled, reflective...) most annealed and toughened glasses can be incorporated in laminates. The combination to be considered depends essentially on the function the glazing must fulfill: simple security against accidental shock, reinforced protection of property against vandalism, bullet resistance, etc.

THICKNESS AND SIZE:

The most common thicknesses are from 6 to 40 mm, but any inquiry may be considered.

The maximum dimensions are those of the glass panes, i.e. 3200 mm x 2250 mm.

PERFORMANCE:

Solar: In addition to meeting requirements for a safety glazing material, laminated glass possesses other unique properties which make it an excellent product for commercial buildings. Laminated glass provides the greatest reduction of ultraviolet light transmission of any commercial glass product. When combined with two layers of float glass, less than 1% of UV is transmitted.

Sound: Laminated glass reduces noise transmission due to the sound dampening characteristics of the interlayer. While glass is inherently a poor acoustical performer, higher performance levels can be achieved with the use of laminated glass alone or combined with additional glass plies to form a sealed insulating glass unit.